

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Python For Data Science Practical

Practical – II

Roll No.: S021	Name: LODHI IQRA
Class: SYBSc IT	Subject: DATA STRUCTURE
Date of Assignment: 22/07/2026	Date/Time of Submission: 22/07/2026

TUPLE PROGRAM

2a. Write a python code to create nested tuples.

CODE:

```
student = ("IQRA", 21, ("BscIT", "SEMESTER III"), 90)

print("Name : ", student[0])
print("Roll no : ", student[1])
print("Department : ", student[2][0])
print("Semester : ", student[2][1])
print("Percentage : ", student[-1])
print("IQRA S021")
```

OUTPUT:

```
===== RESTART: C:/Users/IT-32/
Name : IQRA
Roll no : 21
Department : BscIT
Semester : SEMESTER III
Percentage : 90
IQRA S021
```

2b. Write a python code to sort the nested tuple using the sorted () function.

CODE:

```
data = [(5,8),(3,4),(8,9)]
sorted_data = sorted(data, key = lambda x : x[1])
print(sorted_data)

print("IQRA S021")
```

OUTPUT:

```
===== RESTART: C:/Users/
[(3, 4), (5, 8), (8, 9)]
IQRA S021
```

2c. Write a python code to copy or clone a list.

CODE:

```
list1 = [10,20,30,40,50,60]
list2 = list1.copy()
print("Original list : ",list1)
print("Copied list : ",list2)
print("IQRA S021")
```

OUTPUT:

```
===== RESTART: C:/Users/IT-32/Documents/
Original list : [10, 20, 30, 40, 50, 60]
Copied list : [10, 20, 30, 40, 50, 60]
IQRA S021
> |
```

2d. Write a python code to check the immutability property of python tuples.

CODE:

```
File Edit Format Run Options Window Help
fruit = ("CHERRY", "APPLE", "LITCHI")
fruit[2] = "STRAWBERRY"
print(fruit)

print("IQRA S021")
```

OUTPUT:

```
===== RESTART: C:/Users/IT-32/Documents/PYTHON/practical2.py =====  
Traceback (most recent call last):  
  File "C:/Users/IT-32/Documents/PYTHON/practical2.py", line 2, in <module>  
    fruit[2] = "STRAWBERRY"  
TypeError: 'tuple' object does not support item assignment  
|
```